Z + jets Run 3: data/MC before and after the fixes

What changed, how we found it, and what is still open

Small run: 20 files per sample, four campaigns (2022 C–D, 2022 E–G, 2023 C, 2023 D)

29 October 2026

Three fixes

Same files and same selection before and after. Only these changed.

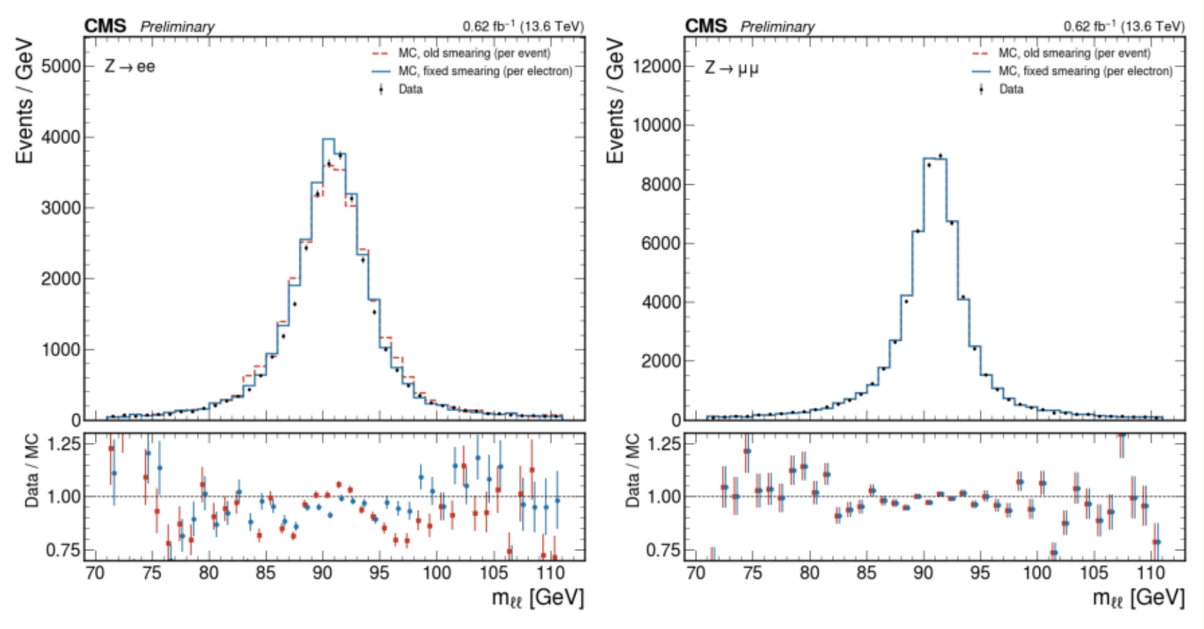
Fix	What was wrong	How we found it
1. Electron smearing	Both electrons got the same random	Z→ee peak wider in MC
	number, so MC Z→ee was too wide	than in data
2. MET	The electrons' own jets were counted	High-pT Z→ee data lost
	in the MET (fake MET in Z→ee)	at the MET < 50 cut
3. Z pT weights	The DY generator gets the Z pT	Wiggle in Z pT before
	shape wrong	any cut, both channels

Fixes 1 and 2 are bugs. Fix 3 is a correction we derive.

Fix 1 and 2 change shared framework code, in our fork only. Fix 2 follows the JME type-1 MET recipe, which the framework applied only for 2024–25 samples. Fix 3 weights DY MC to Z→μμ data and is checked on Z→ee.

Fix 1: Z→ee mass width (2023 D shown)

Red: MC before. Blue: MC after. Muons are not touched by this fix.



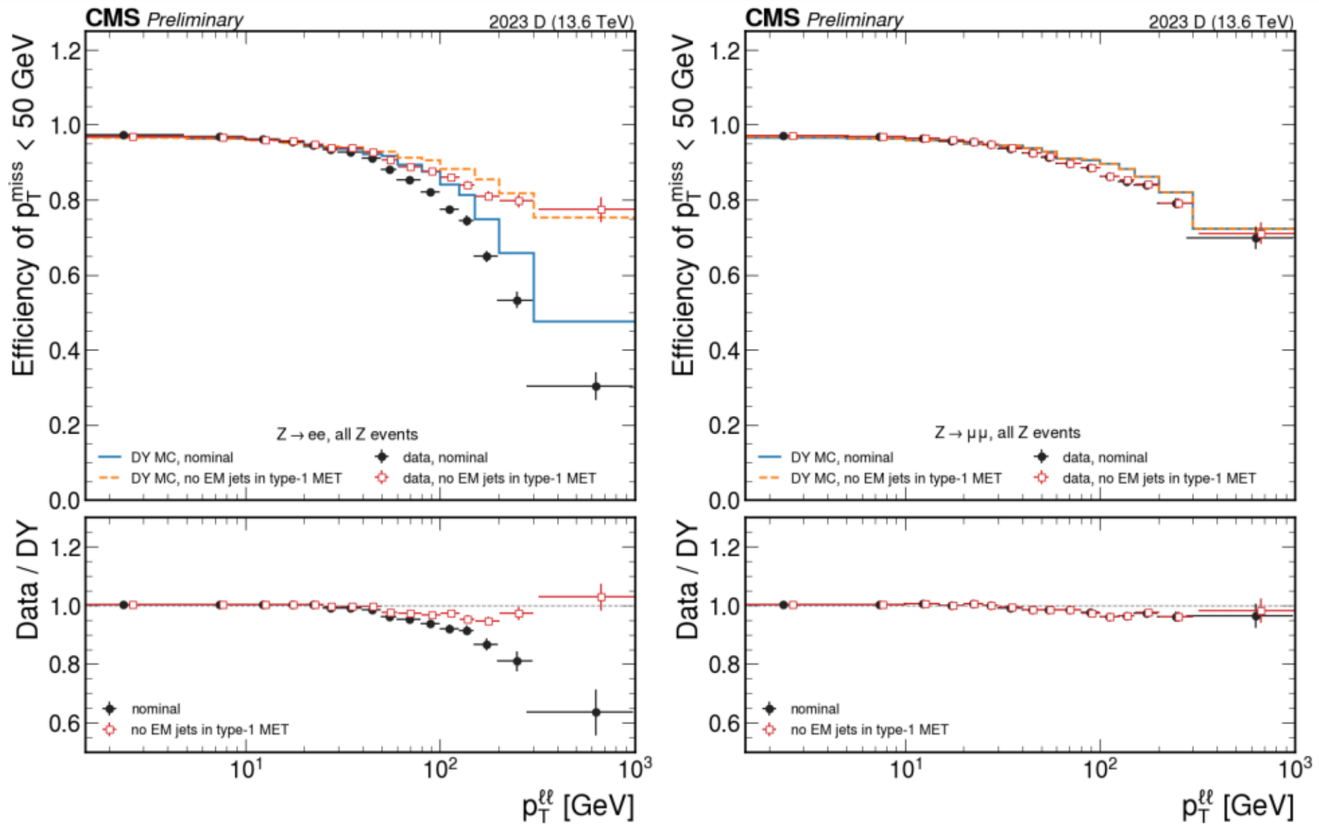
Width data/MC (Z→ee)	2022 C–D	2022 E–G	2023 C	2023 D
before	0.95	0.94	0.94	0.90
after	0.99	1.00	1.00	0.97

Width fixed. The peak position did not move (this fix only changes the width).

Data peak is 0.03–0.08 GeV (< 0.1%) above MC for electrons, same as before. That is inside the EGM energy-scale uncertainty (0.05–0.36%), so it is a systematic, not a bug.

Fix 2: why high-pT electrons were lost (2023 D)

Fraction of Z events passing MET < 50 GeV, before the jet cut. Filled: now. Open: electrons' jets left out of the MET.



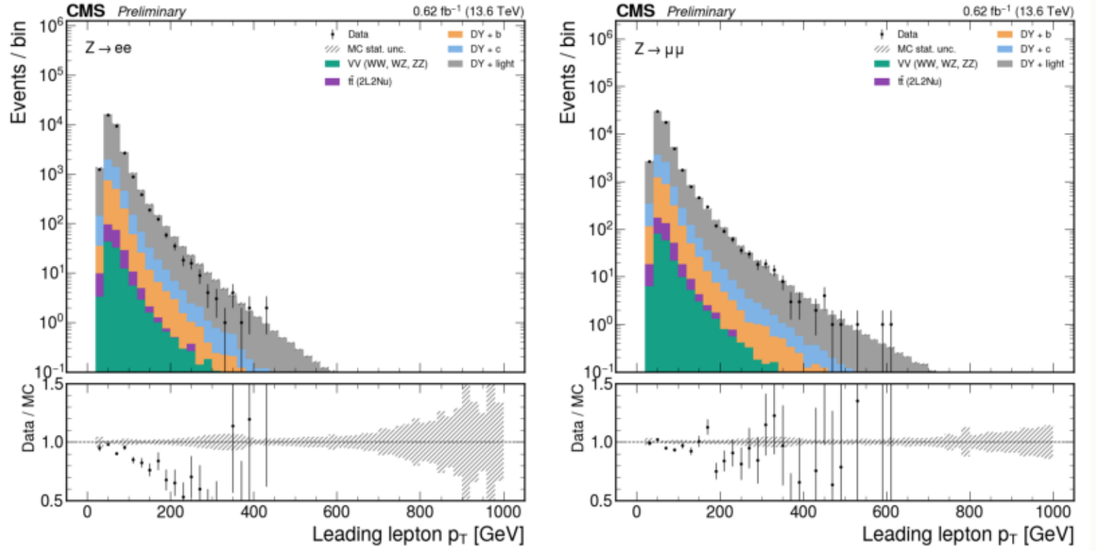
The electron's own jet made fake MET, more in data than in MC.

Z→ee data/MC of the MET cut: 0.81 at Z pT 200–300 GeV, 0.64 above 300. Without those jets: 0.97 and 1.03, the same as Z→μμ. Muons do not change, as expected.

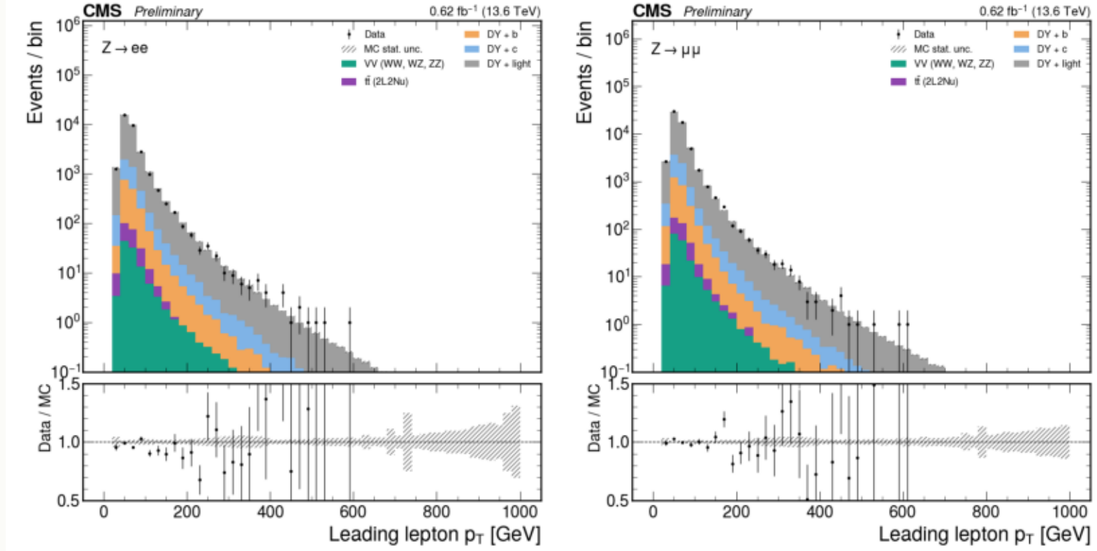
Leading-lepton pT in 2023 D: before and after

Same plot as slide 14 of the main deck. Left: before. Right: after.

Before



After

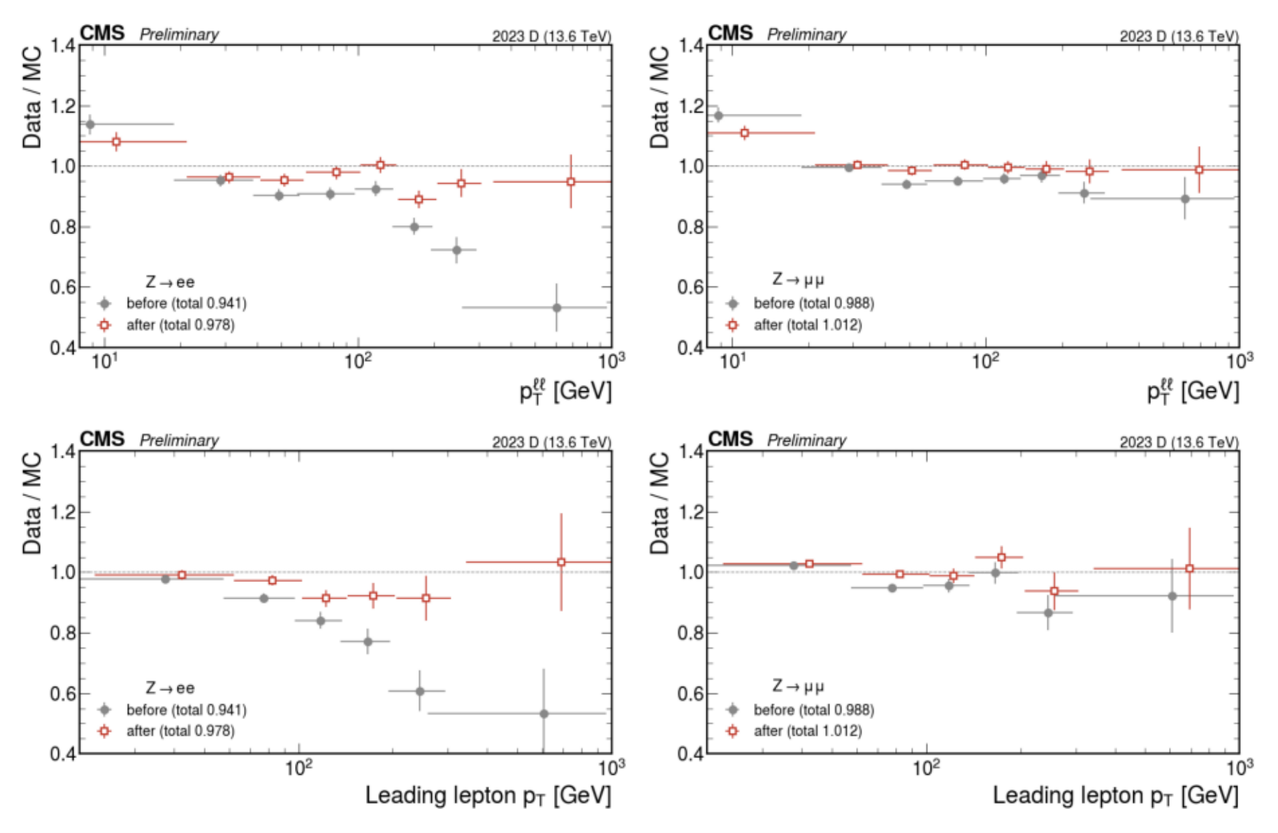


The Z→ee drop at high pT is gone.

Z→ee data/MC at leading pT 200–300 GeV: 0.61 → 0.91. Above 300 GeV: 0.53 → 1.03. Z→μμ barely changes.

Data/MC before and after: 2023 D

Full selection. Top: Z pT. Bottom: leading-lepton pT. Grey: before. Red: after. Wide bins, errors = data + MC statistics.

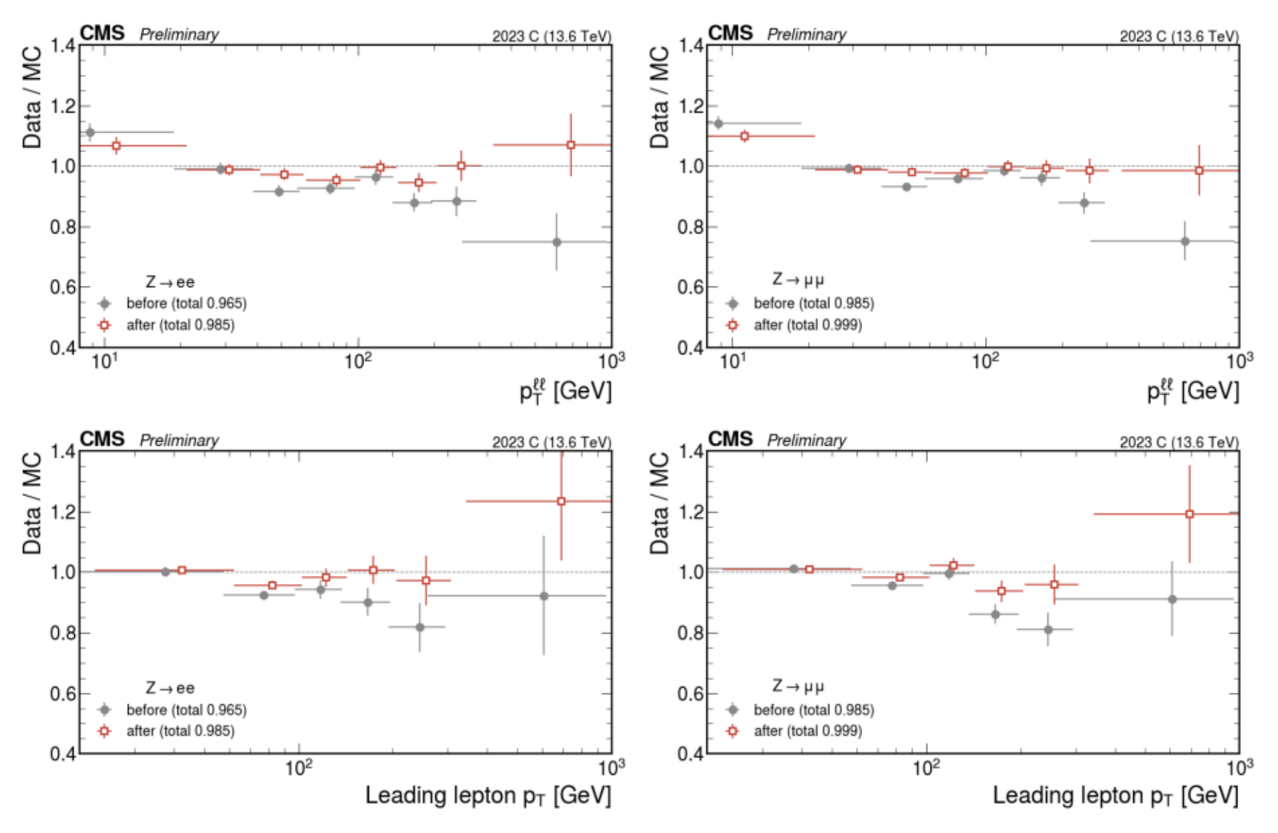


2023 D: clear improvement in Z→ee

Z→ee at Z pT 200–300 GeV: 0.72 → 0.94, above 300: 0.53 → 0.95. Z→μμ gets a little flatter. Not fixed: the first bin (Z pT < 20 GeV) is still 1.08–1.11.

Data/MC before and after: 2023 C

Full selection. Top: Z pT. Bottom: leading-lepton pT. Grey: before. Red: after. Wide bins, errors = data + MC statistics.

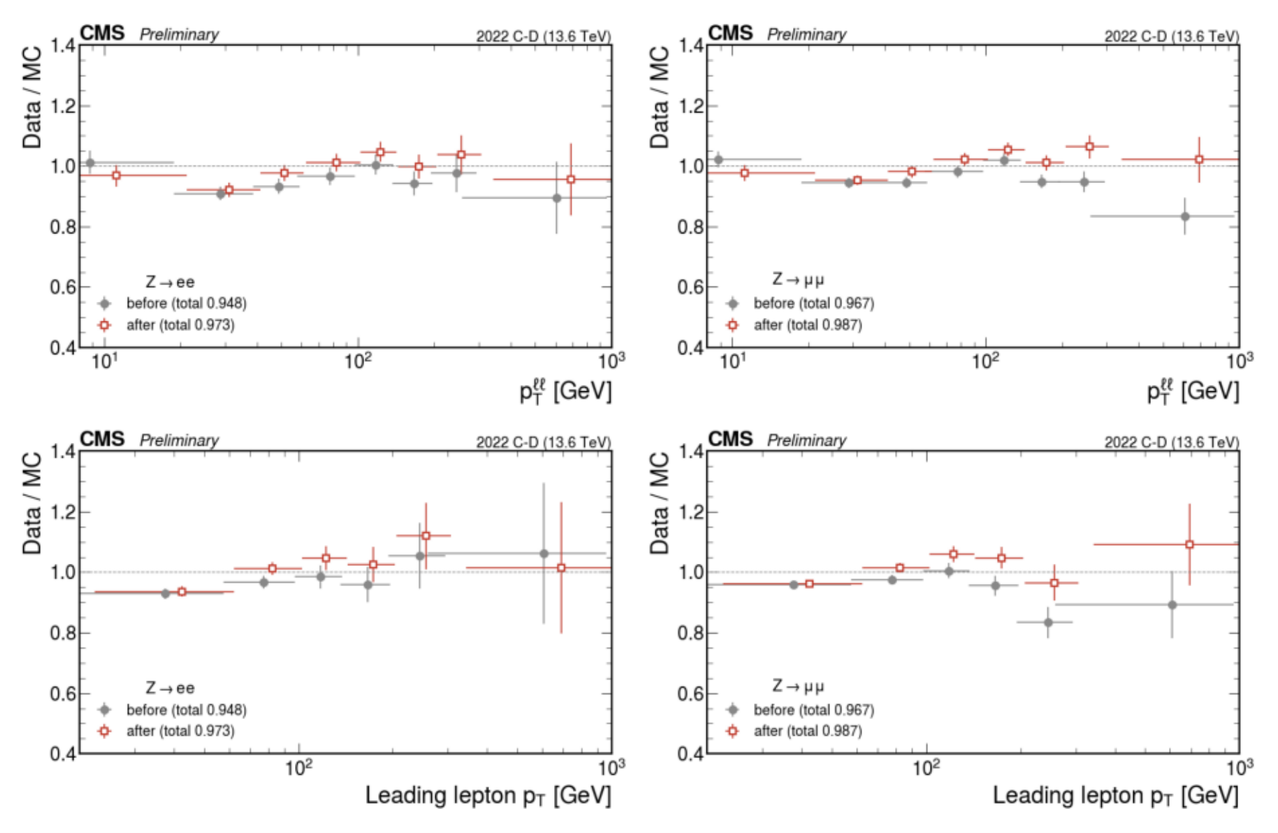


2023 C: better at high pT, within errors elsewhere

Z pT above 200 GeV: 0.75–0.88 → 0.98–1.07 in both channels (each bin 1–2σ). Not fixed: first bin still 1.07–1.10.

Data/MC before and after: 2022 C–D

Full selection. Top: Z pT. Bottom: leading-lepton pT. Grey: before. Red: after. Wide bins, errors = data + MC statistics.

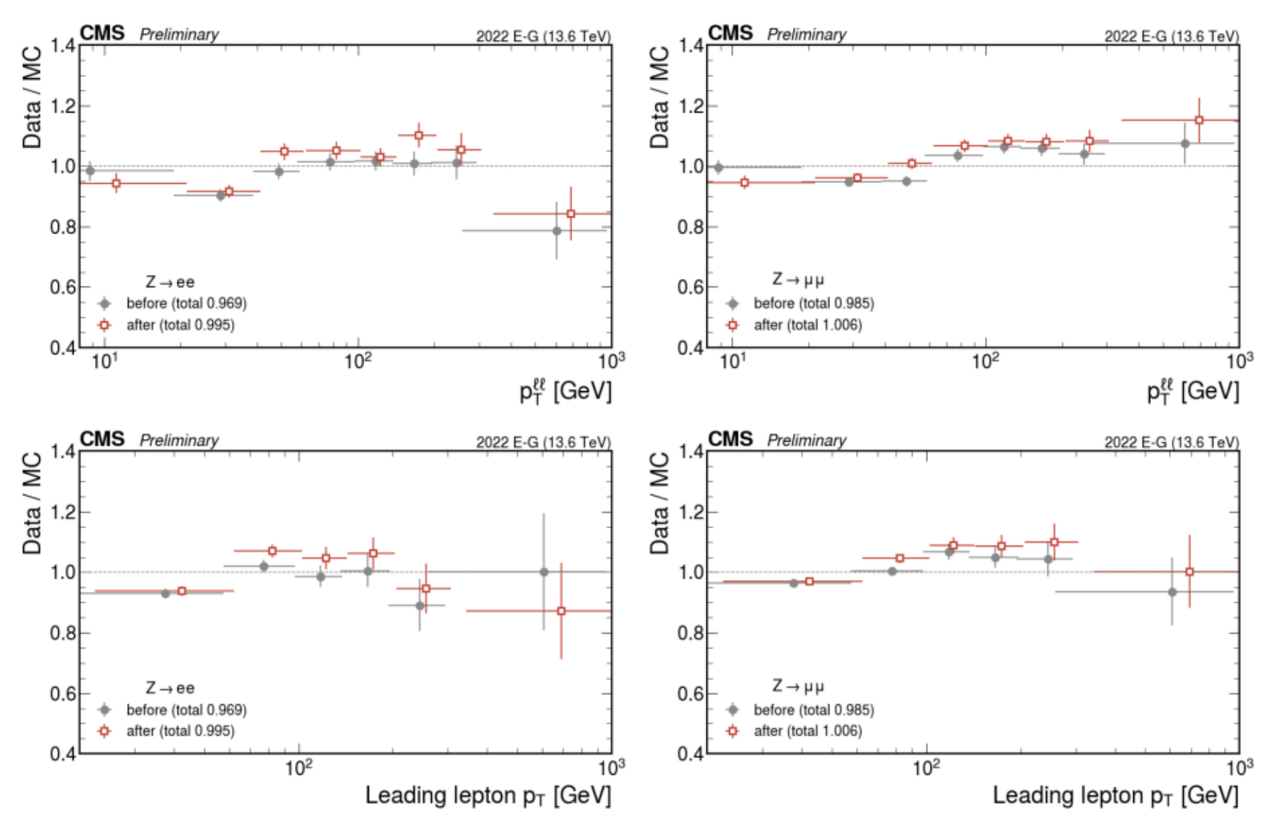


2022 C–D: level up 2%, shape not better

Totals 0.948/0.967 → 0.973/0.987. Shape: low at 20–40 GeV (0.92–0.95), high at 100–140 GeV (1.05), before and after. Likely the 2022 jet excess (page 11).

Data/MC before and after: 2022 E–G

Full selection. Top: Z pT. Bottom: leading-lepton pT. Grey: before. Red: after. Wide bins, errors = data + MC statistics.

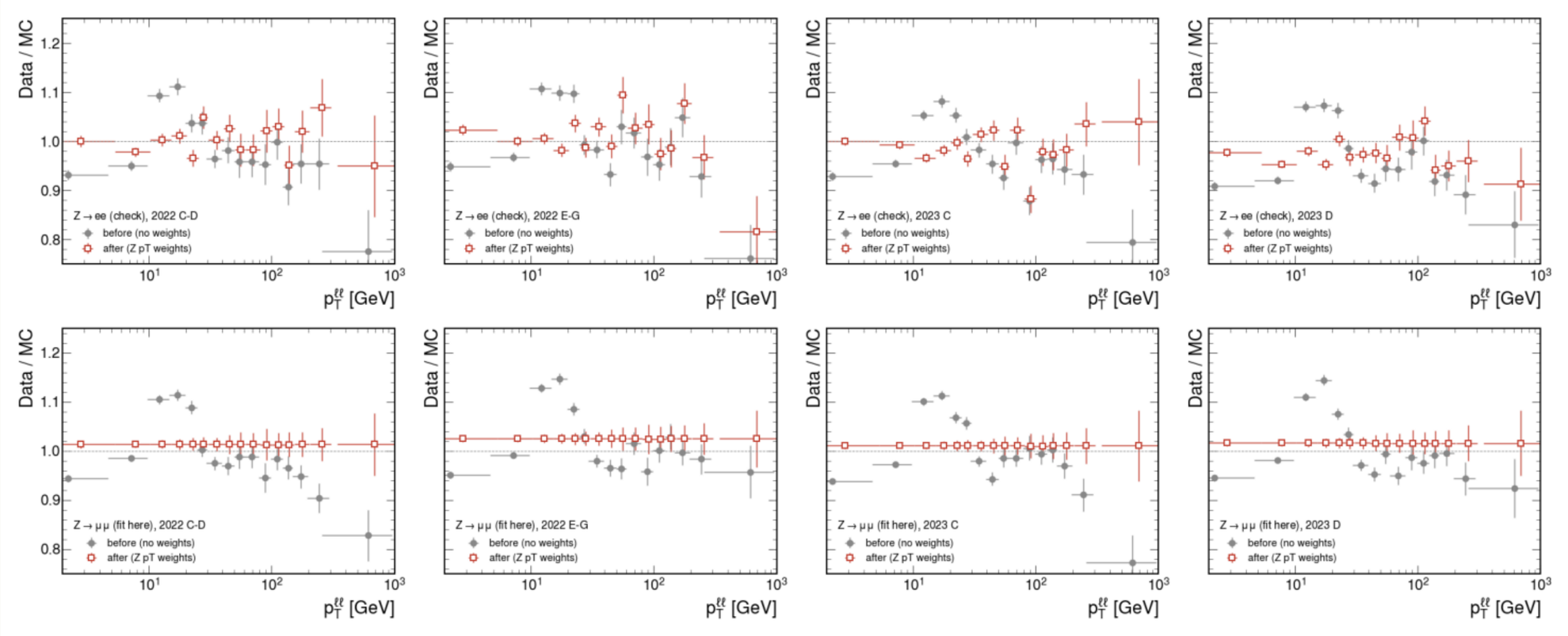


2022 E–G: level up 2%, shape slightly worse

Totals 0.969/0.985 → 0.995/1.006. Z→μμ at 60–300 GeV was 1.03–1.06, now 1.07–1.08. Low at 20–40 GeV as before. Same 2022 jet issue (page 11).

Fix 3: Z pT weights, checked on Z→ee

All Z events, before the MET and jet cuts. Weights come from Z→μμ (bottom row, flat by construction). Z→ee (top row) is the independent check.

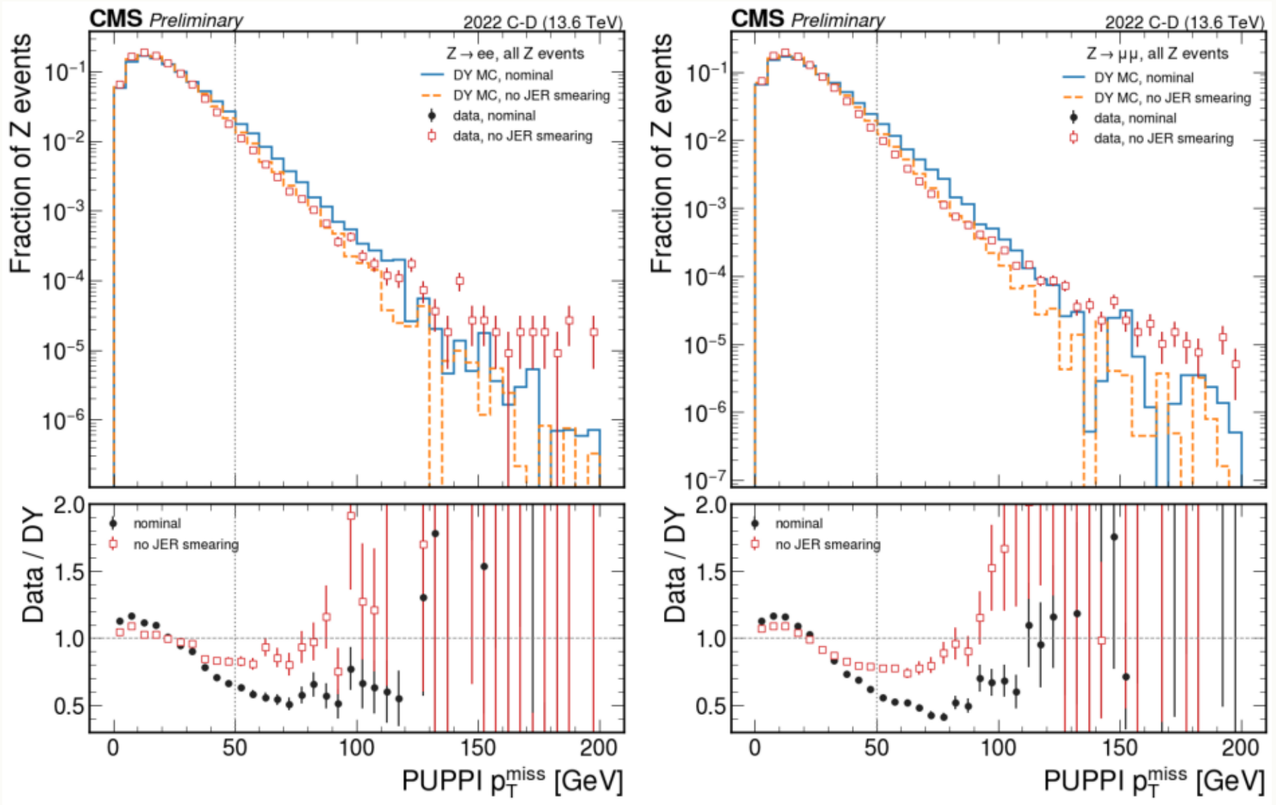


The weights remove the generator wiggle in Z→ee too.

Bin-to-bin spread of Z→ee data/MC: 7–8% → 3–6%. Totals unchanged. After the jet cut the effect is smaller, and the first Z pT bin is still high (pages 6–7).

2022: extra MET and jets in MC (open)

MET of all Z events, 2022 C–D. Filled: now. Open: MC without jet energy smearing (JER). Data is the same in both.



Jet smearing makes most of the extra MET, but not the extra jets.

Data/MC above 50 GeV: about 0.5 → 0.8 without JER. MC events with ≥1 jet: 5% too many → 3%. Turning JER off is only a test.
Next: check the 2022 JER inputs (2023 uses newer ones and is fine).

Summary

Data/MC, Z→ee / Z→μμ	before	after	shape
2022 C–D	0.948 / 0.967	0.973 / 0.987	**not better**
2022 E–G	0.969 / 0.985	0.995 / 1.006	**slightly worse**
2023 C	0.965 / 0.985	0.985 / 0.999	**better at high pT**
2023 D	0.941 / 0.988	0.978 / 1.012	**much better (Z→ee)**

Done: two real bugs fixed, and the big 2023 D electron slope is gone.

Z→ee width matches data. High-pT electrons are no longer lost to fake MET. All totals are now 0.97–1.01.

Open

2022 jets and MET (JER inputs). First Z pT bin 7–11% high in 2023. 2023 D electrons 3% below muons (ask EGM). Z→ee peak offset < 0.1% (EGM scale systematic). 20 files are too few for few-% shapes: full condor run next.